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Data, Justice, and Society

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### Data Portrait

My goal for the data portrait was to investigate my residence hall's floor lounge. I was specifically interested in how the lounge reflects community traits, needs, and preferences. I drew inspiration from DuBois, who immersed himself in his research by living in and actively engaging with the community he studied. DuBois worked at Atlanta University while researching the African American community in Atlanta (Du Bois, 2018). I mirrored this approach by studying my floor lounge, where I frequently spend time and interact with others. As an engaged community member, I felt motivated to capture data that considers and values members' perspectives and values. I chose observation as my data collection method to limit volunteer bias and obtain more "raw" data. This method allowed me to closely observe lounge occupants without actively engaging or interfering.

It is essential to consider the limitations of observational data collection. To gather sufficient data, I collected seven and a half hours' worth. This process was labor-intensive and inefficient. Additionally, human error is a serious consideration. Maintaining focus while recording data on multiple subjects simultaneously over an extended period was challenging. Convenience bias was another significant limitation. I collected data only from the floor of the residence hall I live in: floor 10 of Dykstra Hall. This was my most efficient option, but I could have gathered data from a busier lounge, allowing for greater representation and a more extensive dataset. Because I could not remain in the lounge for seven and a half hours continuously, I collected data over three days. This is a significant external factor, as daily

variations in activity could have impacted the data. Most notably, Tuesday, November 5, was election day. This event may have influenced variables like mood; people may be more prone to mood extremes on election day.

I created multiple data visualizations to tackle the broad challenge of describing lounge activity and relating it to the community's needs and preferences. I chose line graphs for noise level and number of occupants to present change over time. Spikes, dips, and plateaus are easy to identify. I used stacked bar charts when comparing different variables. This type of data visualization makes it easier to identify relationships between variables. To show the distribution of single-variable data, I constructed pie charts. Stacked bars and pie slices allow viewers to compare proportions to each other and to the whole.

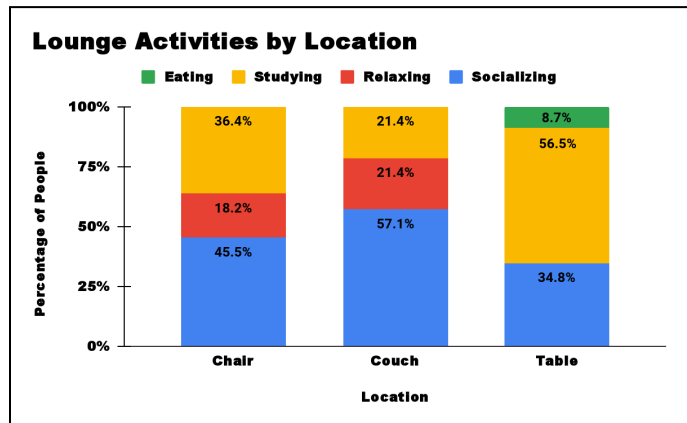
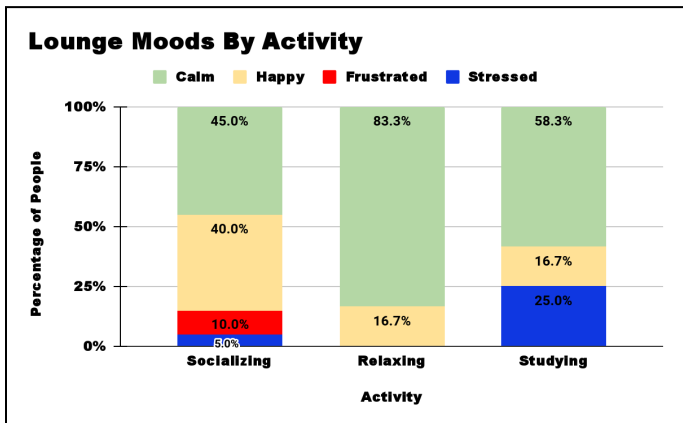
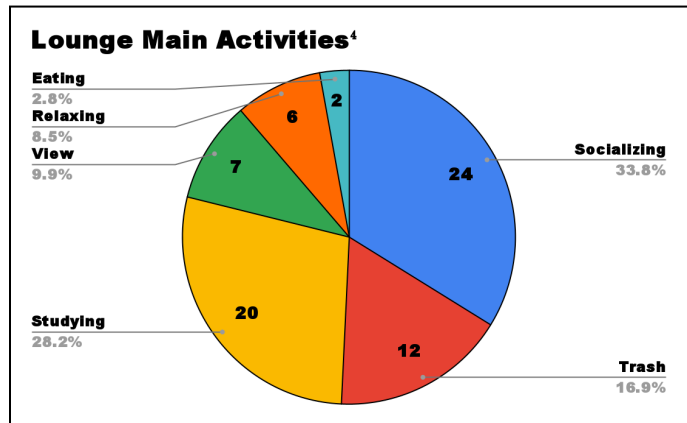
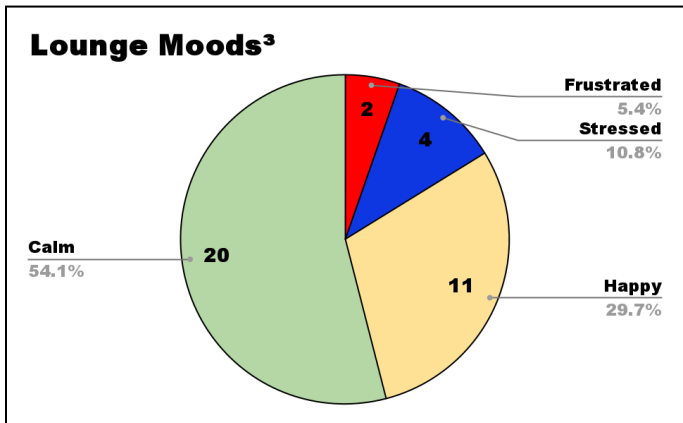
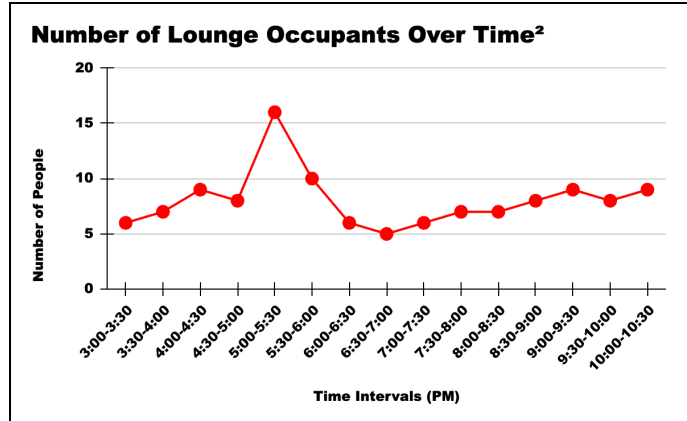
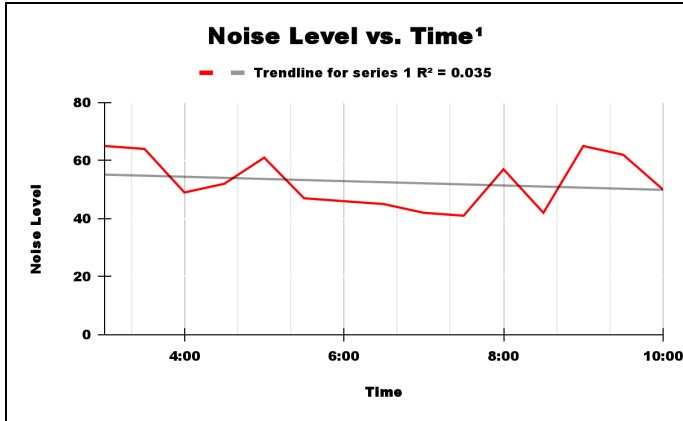
I labeled the stacked bars with percentages to optimize interpretability and accessibility. A major obstacle to structuring the data was defining categories and assigning subjects to only one category. Sorting people's activities was challenging, considering many people multitasked or quickly switched between activities. This obstacle revealed the inherent issues with fixed categories and raised concerns about how they might oversimplify the complex stories behind the data.

I used DuBois' data portrait to inspire effective and impactful data visualizations. DuBois carefully curated powerful, diverse visualizations through strategic stylistic choices, with each design element serving a specific purpose. For instance, DuBois utilizes colors to convey messages. Specifically, in plate 44, DuBois intentionally uses black slices to represent Black workers and white slices to represent white workers. The design illustrates a sharp contrast between the races, relaying a more relevant message about the division within society (Du Bois, 2018). When designing my data portrait, I experimented with color to highlight different aspects

of the community. I assigned colors that aligned with the moods they represented. I used vibrant, stimulating colors for negative emotions and muted tones for positive emotions. Through this approach, I aimed to evoke a reaction in viewers. The emotional associations of colors affect how we perceive data.

The lounge data reveals key trends and conveys a deeper message about its role in the community. The activity pie chart shows that the lounge is a multifunctional space that suits students' needs and activities. With no correlation between time and noise level, the lounge's atmosphere seems driven more by the activities and interactions within it than by predictable temporal patterns. The mood chart shows that about 84% of subjects experienced a positive mood while in the lounge, suggesting it is a supportive, beneficial space for students. Despite its limitations, this data portrait still captures a meaningful story. If the study were refined, it could inform policy. For example, the university could shift the lounge layout to better suit students' needs, such as adding more tables so students have more study space. DuBois' data portrait represented a powerful convergence of visual art and social science used to demonstrate African American progress (Du Bois, 2018). Similarly, this data portrait depicts a community and its traits and behaviors, conveying the lounge's purpose in students' lives.

## Data Visualization



## Footnotes

1. Noise level was tracked using the sound level meter on Decibel X, an app that measures noise levels using air pressure. Noise level is measured in decibels. This was tracked over three days and compiled to provide a composite view.
2. The number of people includes those already in the lounge at the start of each interval and those who entered during the interval. This was tracked over three days and compiled to provide a composite view.
3. The mood of a lounge occupant was only recorded if they were in the lounge for longer than 5 minutes.
4. Only one activity was recorded per lounge occupant. To determine their main activity, I chose the activity they spent the most time doing. “Trash” indicates an occupant throwing away trash in the lounge bins, and “view” signifies an occupant watching the view out the lounge window.

## Works Cited

Du Bois, W. E. B. (2018). *W.E.B. Du Bois's data portraits: Visualizing Black America: the color line at the turn of the twentieth century* (W. Battle-Baptiste & B. Rusert, Eds.; First edition). The W.E.B. Du Bois Center at the University of Massachusetts Amherst.